

Priscilla Wambui Baiya

AI/ MACHINE LEARNING ENGINEER | DATA SCIENTIST

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PERSONAL STATEMENT

MSc Computer Science (AI) graduate who builds and evaluates AI systems from data through to deployment. Built a retrieval-augmented assistant end to end, raising recall@5 from 0.80 to 0.90 through structured testing. As a Data Scientist at Costplan Group, took a data and AI proof of concept from problem definition to director sign-off. Seeking a graduate ML, AI engineering or data science role.

EDUCATION

MSc Computer Science (Artificial Intelligence), University of Kent

2024 – 2026

- Distinction (expected). Modules: Machine Learning Algorithms, Deep Learning, Data Mining.
- Built a three-stage generative pipeline (Stable Diffusion XL, AgeTransGAN, CodeFormer) for synthetic facial age-estimation data; evaluated failure modes and demographic fairness (FairFace, InsightFace, FID).
- Follow-up study, with supervisor, of GPT-Image-1, Midjourney and Stable Diffusion: 43–49% ethnicity prediction error in all three, suggesting the FairFace evaluator is itself biased. Research paper: [GitHub](#).

BSc Business and IT, Jomo Kenyatta University

2018 – 2022

- 2:1 (Upper Second-Class Honours).

SKILLS

- **Machine Learning & AI:** generative AI (diffusion models, GANs), LLM tooling and RAG (Groq API), model and fairness/bias evaluation, PyTorch, TensorFlow, scikit-learn.
- **Data & Databases:** Pandas, NumPy, data cleaning & validation, Matplotlib, Seaborn, Power BI, DAX (basic); ETL/ELT pipelines, star schema and database design (SQL Server, Azure SQL, SQLAlchemy).
- **Engineering & Testing:** Python, SQL, FastAPI, Git, Docker Compose, pytest, GitHub Actions CI, mypy, ruff, pre-commit.

RELEVANT PROJECTS

- **RAG construction assistant:** Built a retrieval-augmented assistant answering construction and site-safety questions from UK Building Regulations and HSE guidance, citing the exact clause behind every answer.
- Designed a hybrid BM25 + dense (Chroma, BGE-small) retrieval pipeline with rank fusion and a labelled evaluation harness; fixed a source-imbalance failure (one document dominating results) with capped, backfilled ranking, raising recall@5 from 0.80 to 0.90 on 25 hand-labelled questions.
- Deployed as a Streamlit app and CLI with Groq-hosted LLMs constrained to cite only retrieved passages; includes an evaluation pipeline and test suite. Code: [GitHub](#)
- **Berlin S-Bahn delay prediction:** Framed delays (6+ minutes) in 131,771 synthetic Kaggle trips as a classification task with 32:1 class imbalance, so accuracy was avoided as a metric.
- Validated data into SQLite and Parquet; tests guard against leakage and bad temporal splits. Code: [GitHub](#)

EMPLOYMENT

Data Scientist, Costplan Group

Sep 2025 – Sep 2026

- Consolidated cost data scattered across project spreadsheets into a proof-of-concept SQL Server / Azure SQL database and ETL pipeline, with validated Excel ingestion, batch tracking and curated reporting views; containerised with Docker Compose and tested via GitHub Actions CI.
- Prototyped an LLM natural-language query assistant (Groq), restricted to read-only, single-statement queries, so non-technical stakeholders could query data directly.
- Prototyped AI-assisted report generation tested so LLM-drafted narrative stayed grounded in database figures; cut drafting from hours to under 30 minutes.
- Built Power BI dashboards; directors signed off the proof of concept as a two-phase project.

Project Coordinator, AfroCloud Technologies

Dec 2023 – Aug 2024

- Managed three 6-month client projects end-to-end with eLearning developers and NGO partners (EMS, DSW).